

IoT Sensor Data Schema

Purpose: Store and track environmental sensor readings, errors, and startup events from deployed sensor boxes.

nu_sensors

Field	Description
deployment_id	Unique identifier for each sensor device + location. (Primary key)
box_id	Identifier for each sensor device.
location_id	Identifier for sensors within a specific area with shared analytics.
coreid	Particle device core identifier.
address	Human-readable label for the sensor location.
latitude	GPS latitude coordinate.
longitude	GPS longitude coordinate.
install_datetime	When the sensor was installed (UTC format).
is_active	Whether the sensor is currently active.
created_at	Record creation timestamp (UTC format).

nu_readings

Field	Description
id	Auto-incrementing record ID (Primary Key).
deployment_id	References sensors.deployment_id .
box_id	Identifier for each sensor device.
timestamp	When the reading was taken (UTC format).
temperature	Temperature measurement (°F).
humidity	Humidity percentage (%).
noise	Noise level measurement (db).

heat_index	Calculated heat index (°F).
source_blob	Original data source reference (blob file name).
created_at	Record creation timestamp (UTC format).
quality_ok	Boolean value to manually flag quality issues for each record.

nu_errors

Field	Description
id	Auto-incrementing record ID (Primary Key)
deployment_id	References sensors.deployment_id .
box_id	Identifier for each sensor device.
timestamp	When the error occurred (UTC format).
error_code	Specific error identifier (e.g., 'E1')
source_blob	Original data source reference
created_at	Record creation timestamp (UTC format).

nu_startup

Field	Description
id	Auto-incrementing record ID (Primary Key)
deployment_id	References sensors.deployment_id .
box_id	Identifier for each sensor device.
timestamp	When the startup occurred (UTC format).
source_blob	Original data source reference
created_at	Record creation timestamp (UTC format).